

BridgeGuard



BridgeGuard



Field of Innovation	Internet of Things (IoT), Artificial Intelligence (AI), Smart Infrastructure & Structural Health Monitoring
Name of Team Leader/Applicant:	Yancy Mevada
Email ID of Team Leader/Applicant:	26bca080@charusat.edu.in
Contact No	9664588927
Gender of Team Leader/Applicant	Female
Team size	2
Team Members Name/Innovators	Yancy Mevada, Rutvi Modi
Name of Guide With Employee id	Bhaumik Shah - 1065
Email ID of Guide	bhaumikshah.mca@charusat.ac.in
Guide Contact No	7359739497
Current Status	Ideas

BridgeGuard



The graphic features a large cable-stayed bridge over water. Various sensors are indicated with icons and labels: a Vibration Sensor on the top pylon, a Temperature Sensor on the side of the pylon, a Load Cell Sensor on the bridge deck, a Strain Sensor on a support pier, and a Water Level Sensor in the water near the pier. A Wi-Fi symbol is shown above the bridge's main span.

BRIDGEGUARD

SMART BRIDGE HEALTH
MONITORING SYSTEM
USING IOT & AI

Real-time Monitoring | AI Based Analysis | Early Alerts | Safer Infrastructure



Real-time
Monitoring



AI Based
Analysis



Early
Alerts



Cloud Data
Storage



Safer
Infrastructure

PRESENTATION BY:
YANCY MEVADA, RUTVI MODI

MONITOR TODAY, **PROTECT** TOMORROW

Problem Statement



- **Current Challenges**
 - Aging infrastructure is a critical issue, as many bridges exceed their designed lifespan. Current manual inspection methods are expensive, periodic, labor-intensive, and prone to human error, often missing small structural defects.
- **Consequences of Failure:**
 - Loss of life
 - Traffic disruption
 - Significant repair and economic costs
- **Statistics:**
 - Over 40% of bridges require rehabilitation.
- **Problem:**
 - There is a lack of affordable, continuous, real-time health monitoring systems for bridge infrastructure.

Proposed Solution



- **BridgeGuard: AI-Powered Structural Health Monitoring**
- BridgeGuard utilizes smart IoT sensors to provide continuous, automated structural health monitoring (SHM) for bridges. By leveraging advanced AI, the system proactively detects potential hazards before they escalate into safety risks.
- **Key Capabilities**
 - Comprehensive Diagnostics: Real-time stress monitoring, vibration analysis, and corrosion detection.
 - Predictive Analytics: AI-driven damage forecasting and automated crack detection.
 - Operational Efficiency: Instant alert notifications and a centralized cloud dashboard for seamless oversight.

Product: BRIDGEGUARD



- BridgeGuard is an advanced Structural Health Monitoring (SHM) platform that integrates IoT and AI to provide continuous, real-time safety assessments and early failure warnings for bridge infrastructure.
- **System Architecture**
 - **Hardware Layer**
 - IoT Connectivity: Central Gateway (ESP32/STM32) for data transmission.
 - Sensor Suite: High-precision monitoring including vibration/accelerometers, strain gauges, crack detection, corrosion tracking, and environmental (temperature/humidity) sensors.
 - Power Management: Sustainable operation via solar panels with integrated battery backup.
 - **Software Layer**
 - Intelligence: AI-driven predictive analytics for anomaly detection.
 - Interface: A comprehensive real-time monitoring dashboard and dedicated mobile application.
 - Infrastructure: Scalable cloud-based database and analytics engine.
 - Communication: Automated SMS and email alerting system for critical structural updates.

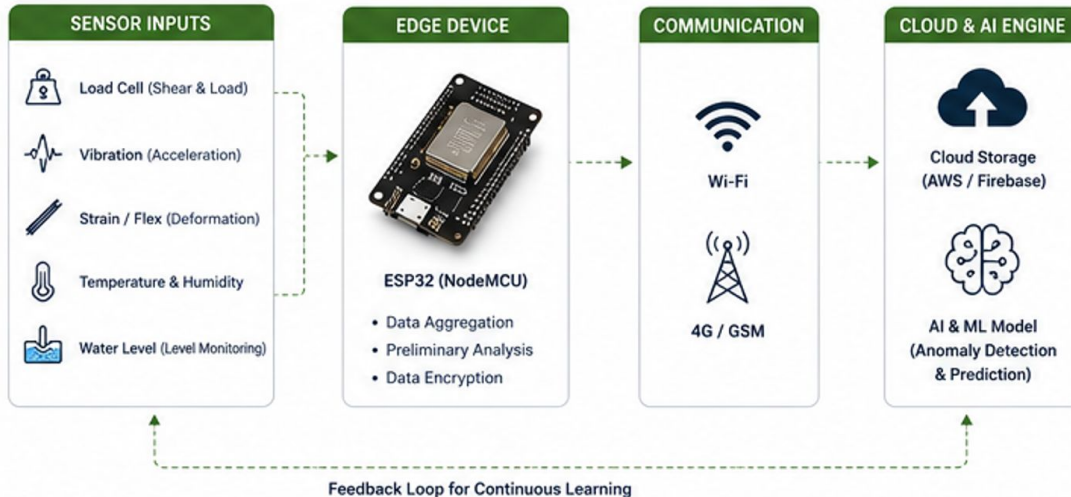
1. SMART BRIDGE MONITORING

Real-time monitoring of critical structural parameters using IoT sensors installed on the bridge.



2. DATA FLOW & EDGE PROCESSING

Sensor data is collected and processed at the edge device (ESP32) for efficient and secure transmission.



3. AI ANALYTICS & PREDICTION

AI/ML models analyze data to detect anomalies, predict trends and assess the health of the bridge.



AI Based Predictive
Maintenance



Early Warnings
& Alerts



Data-driven
Decision Making



Improved Bridge
Lifespan

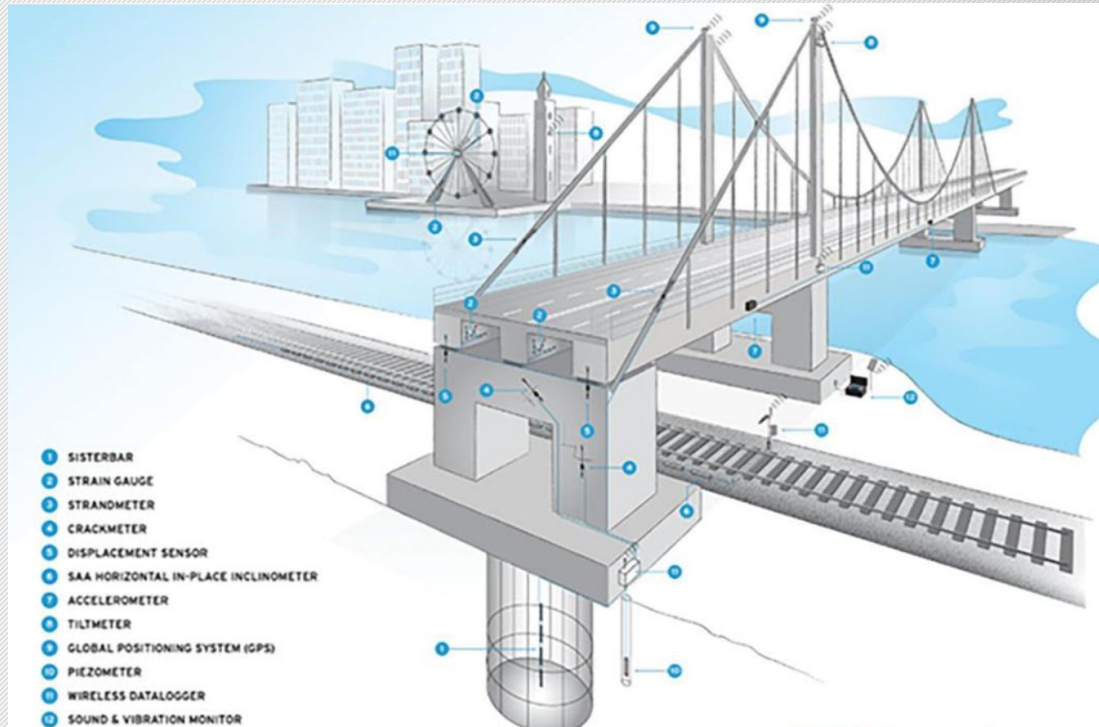
BRIDGEGUARD

4. REAL-TIME DASHBOARD & ALERTS

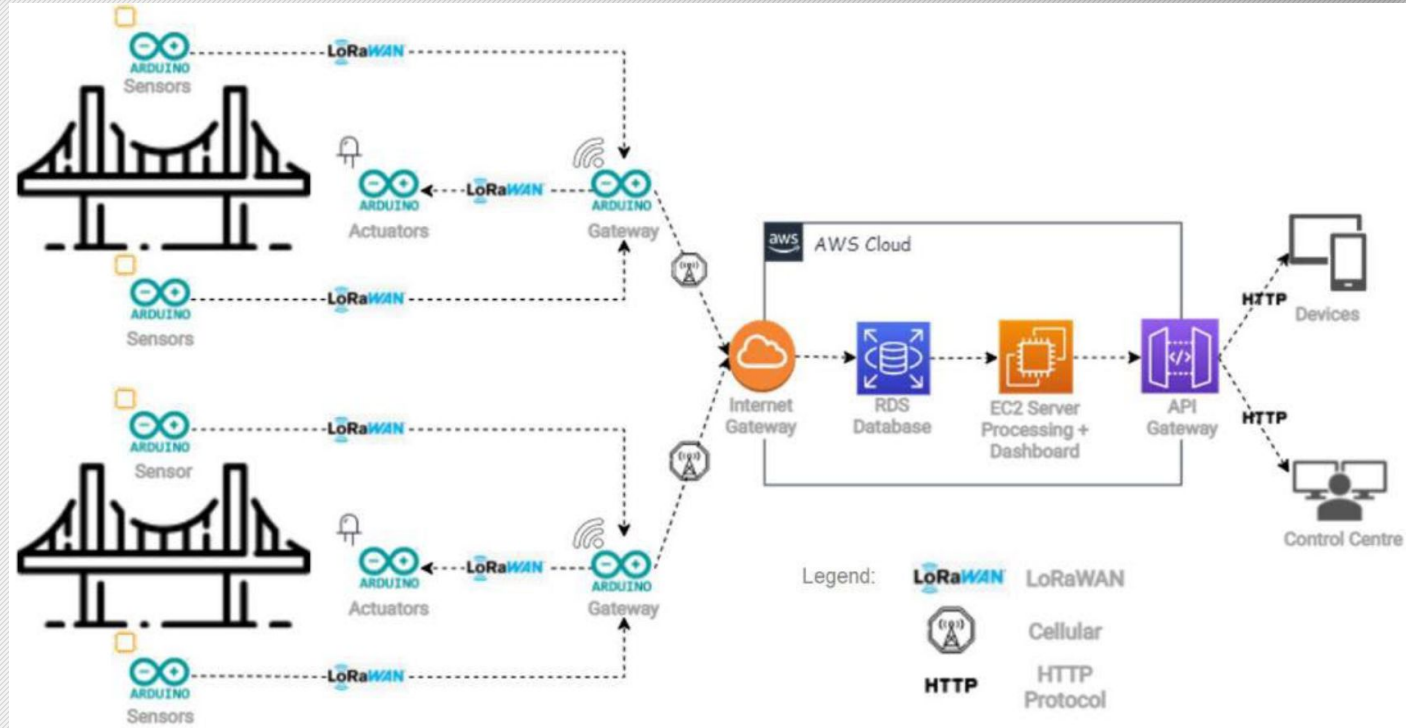
Live data visualization, alerts and notifications for proactive action and safer infrastructure.



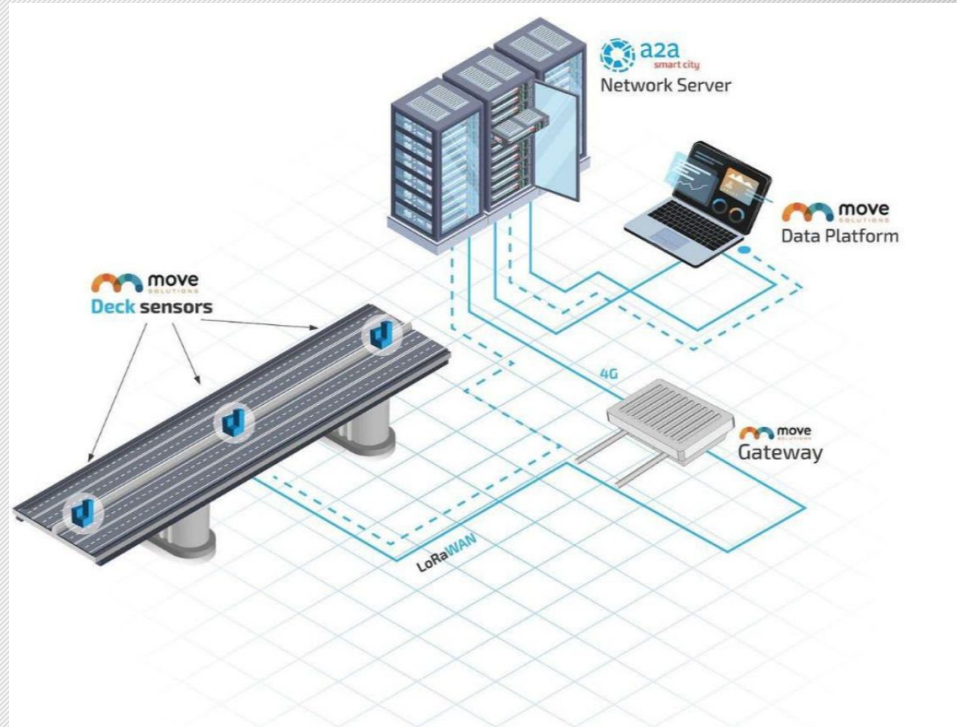
BRIDGEGUARD



BRIDGEGUARD

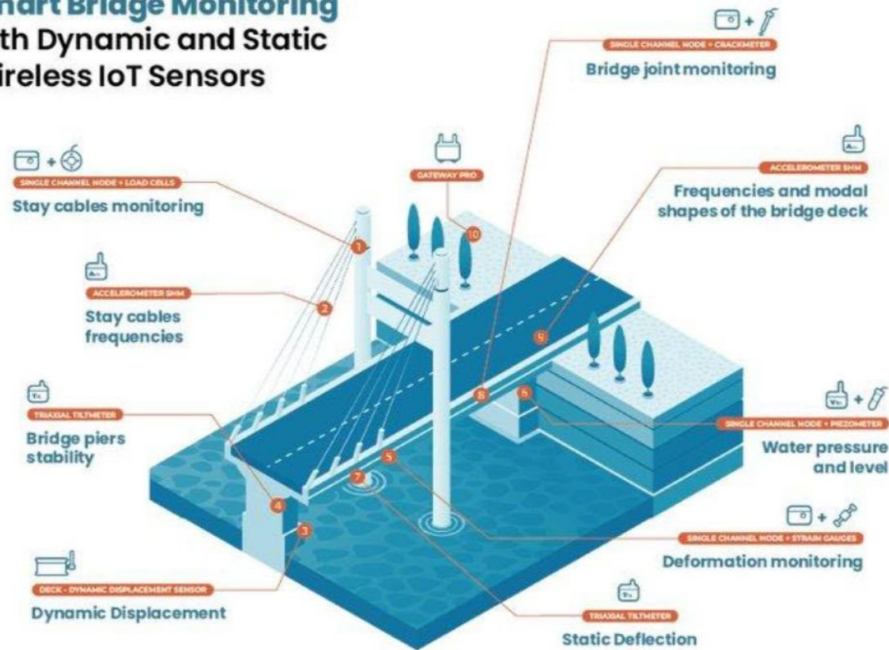


BRIDGEGUARD

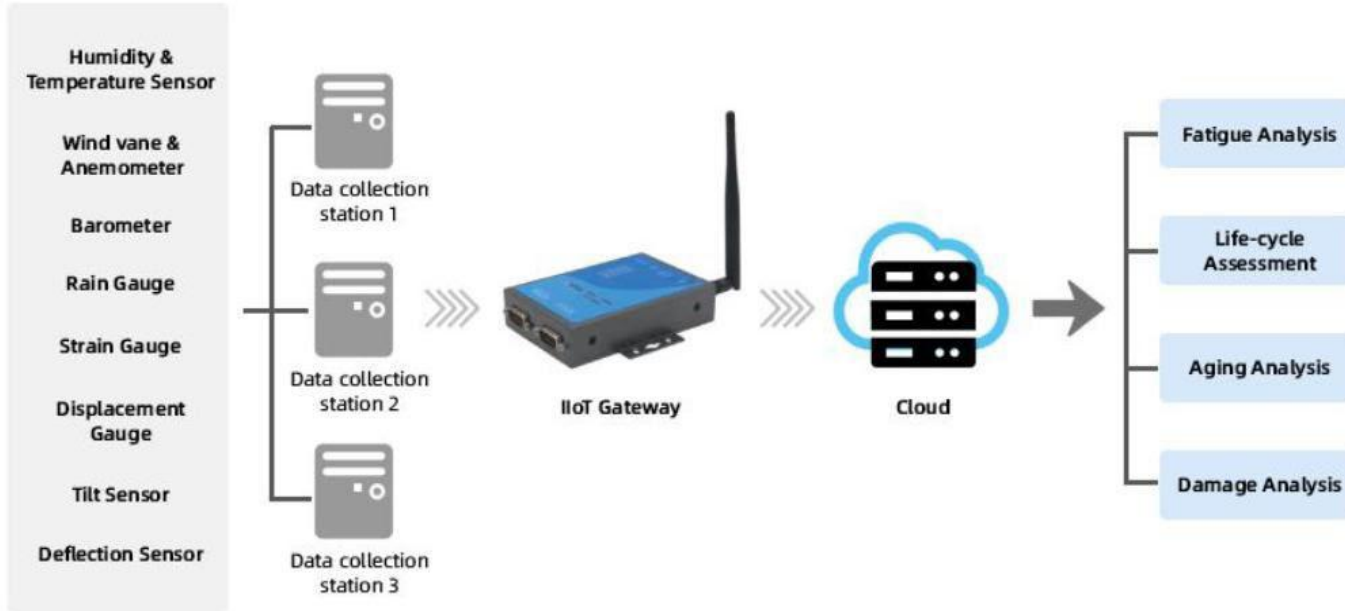


BRIDGEGUARD

Smart Bridge Monitoring with Dynamic and Static Wireless IoT Sensors



BRIDGEGUARD



Unique Selling Proposition

- What sets BridgeGuard apart?
 - Continuous Monitoring: 24/7 real-time oversight.
 - AI-Driven Maintenance: Predictive insights to preempt failures.
 - Proactive Alerts: Early warning system for critical safety.
 - Cost-Efficient: Significant reduction in overall maintenance expenses.
 - Seamless Integration: Designed for easy retrofitting onto existing bridge structures.
 - Sustainable Power: Fully solar-powered operation.
 - Advanced Analytics: Cloud-based data processing and reporting.
 - High Scalability: Versatile application for highways, railway networks, and smart city infrastructure.

Market Opportunity



- **Primary Market Segments**

- Our target audience comprises entities responsible for developing, maintaining, and overseeing large-scale public infrastructure:
 - Government & Public Sector: Public Works Departments, National Highway Authorities, Railway administrations, and Metro Corporations.
 - Urban Development: Smart City project managers and municipal planning bodies.
 - Private & Specialized Services: Major construction firms and infrastructure engineering consultants.

- **Market Growth Drivers**

- The global Structural Health Monitoring (SHM) sector is poised for rapid expansion, fueled by several critical trends:
 - Infrastructure Lifecycle: The urgent need to manage and rehabilitate aging global infrastructure.
 - Technological Integration: The widespread adoption of AI and IoT to enhance monitoring capabilities.
 - Policy & Safety: Increasing public and governmental investments focused on infrastructure safety standards.
 - Urban Modernization: The ongoing development of smart, resilient city initiatives.

Competitors and Competitive Advantage



Global Competitor	Solution They Provide	Their Limitation
Manual Bridge Inspection	Engineers inspect bridges periodically	Time-consuming, costly, and defects may be missed
Sensequake	AI-based structural health monitoring	Expensive and designed for large infrastructure projects
HBK (Hottinger Brüel & Kjær)	High-precision structural monitoring sensors	High installation and maintenance cost
Campbell Scientific	Data acquisition and monitoring systems	Requires complex setup and technical expertise
Geokon	Geotechnical and structural monitoring sensors	Focused on sensor hardware, limited AI-based prediction

Competitors and Competitive Advantage



- **Why Choose BridgeGuard?**
 - **Cost-Effective Innovation:** Provides an affordable IoT-based alternative to traditional, high-cost monitoring systems.
 - **Predictive Maintenance:** Leverages AI to detect potential structural issues before they escalate into critical failures.
 - **Real-Time Monitoring:** Ensures continuous data collection and analysis via a secure cloud infrastructure.
 - **Seamless Installation:** Designed for easy integration with existing bridges, eliminating the need for invasive structural changes.
 - **Scalable Solution:** A versatile system adaptable to bridges of any size—from small crossings to large-scale infrastructure projects.
 - **Optimized Maintenance:** Shifts focus from periodic, calendar-based inspections to cost-saving, condition-based maintenance.
 - **Enhanced Connectivity:** Features a dedicated mobile app that delivers instant alerts, enabling authorities to respond rapidly.
 - **Public Safety First:** Promotes infrastructure longevity and public safety through proactive warnings and constant health monitoring.

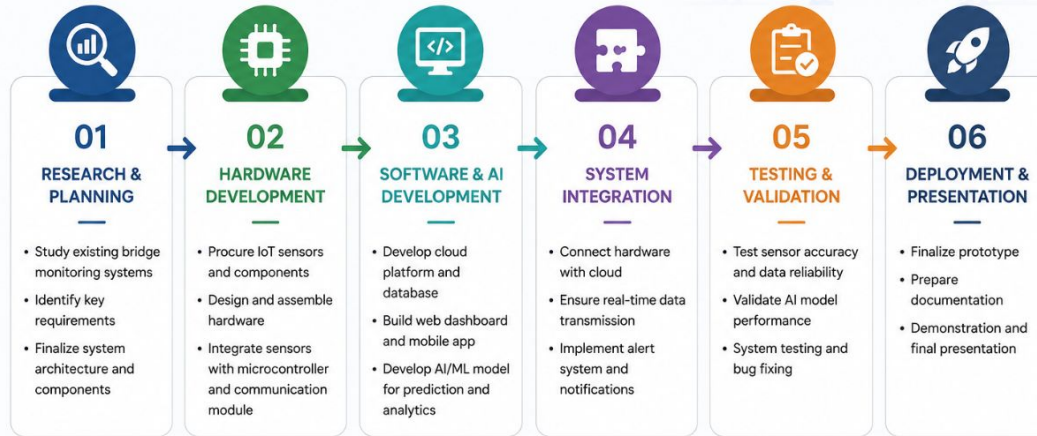
Business Model

- **Target Market & Customer Segments**
 - Government entities (National Highway Authorities & Smart City initiatives)
 - Railway infrastructure departments
 - Major construction and infrastructure development firms
- **Core Value Proposition**
 - **Operational Excellence:** Automated 24/7 structural health monitoring.
 - **Predictive Intelligence:** AI-driven insights for proactive maintenance scheduling.
 - **Risk Mitigation:** Advanced early warning systems to prevent critical incidents.
 - **Economic Impact:** Significant cost reductions coupled with improved public safety outcomes.
- **Revenue Model**
 - Direct hardware procurement and installation services.
 - Recurring revenue via Annual Maintenance Contracts (AMC).
 - Subscription-based cloud access and advanced AI data analytics.
- **Strategic Partnerships**
 - Leading IoT and cloud computing infrastructure providers.
 - Government agencies and key infrastructure stakeholders.
 - Collaboration with academic and technical research institutions.

Work Plan

WORK PLAN

Our step-by-step approach to develop BridgeGuard



OUR GOAL

To develop a reliable, accurate and cost-effective bridge monitoring system that enhances safety and reduces maintenance costs.



Proposed Budget

Item	Estimated Cost (₹)
IoT Hardware (ESP32, GSM/Wi-Fi Module, LCD Display, Power Supply)	₹10,000
Sensors (Load Cell, HX711, Vibration, Strain/Flex, Temperature & Humidity, Water Level)	₹7,000
Prototype Bridge Fabrication & Materials	₹10,000
AI Dashboard & Cloud Services	₹6,000
PCB, Wiring, Connectors & Miscellaneous	₹2000
Documentation & Printing	₹3000
Total Estimated Budget	₹38,000

Your Expectations from CSIC



- To transform BridgeGuard from a prototype into a market-ready solution, we are seeking support from CSIC in the following areas:
 - **Funding & Resource Allocation**
 - **Seed Funding:** Financial assistance for prototype development.
 - **Procurement:** Support for acquiring necessary hardware and testing components.
 - **Infrastructure:** Access to advanced labs, testing facilities, and specialized workspaces for product validation.
 - **Technical & Strategic Mentorship**
 - **Expert Guidance:** Mentorship in IoT, AI, and comprehensive product development.
 - **Performance Optimization:** Technical feedback to refine and improve system capabilities.
 - **Business Development:** Support for refining our business model, preparing investor pitches, and navigating IP and patent landscapes.
 - **Industry Integration & Networking**
 - **Strategic Connections:** Facilitated introductions to government agencies and key infrastructure firms.
 - **Pilot Deployments:** Opportunities to test and validate the system on real-world bridges.
 - **Expected Outcome**
 - With CSIC's collaboration, BridgeGuard will evolve into a scalable, smart infrastructure solution that enhances bridge safety, drives down maintenance costs, and strengthens the foundation for safer smart cities.

Thank you

